

Unsupervised and semi-supervised anomaly detection on the LHC Olympics 2020 R&D dataset

Mithil Hardik Astik¹

¹Department of Physics, BITS Pilani, Hyderabad Campus, Hyderabad, Telangana, India.

✉ e-mail: f20240923@hyderabad.bits-pilani.ac.in

Model-agnostic searches for new physics at the Large Hadron Collider rely on anomaly detection, yet trade-offs between supervision, training contamination and invariant-mass distortion remain poorly quantified. I benchmark unsupervised, weakly supervised and supervised detectors on the LHC Olympics 2020 R&D dijet dataset using five jet-substructure features. Isolation Forest is the strongest learned unsupervised detector ($AUC = 0.807 \pm 0.008$); Deep SAD with 100 labelled anomalies reaches $AUC = 0.888 \pm 0.011$, against 0.9659 for the supervised reference, with $JSD = 0.0066$ and a descriptive local fixed-window diagnostic. Deep SAD retains $AUC = 0.749 \pm 0.061$ on held-out three-prong events. Follow-up extensions test a masked Deep Sets autoencoder on constituent inputs, two external signals, conformal calibration of an ANODE score and background-only closure. The Deep Sets AUCs are 0.498 (nominal two-prong), 0.441 (three-prong), 0.465 ($X \rightarrow YY'$) and 0.507 ($W_{KK} \rightarrow WR \rightarrow 3W$). Sideband conformal p -values are consistent with uniformity ($KS = 0.0043$, $p = 0.179$); background-only closure shows no local $Z \geq 3$ fluctuations for two of three tested scores at a nominal 1% cut. The richer representation did not improve discrimination, and calibration and closure claims remain conditional on simulated background.

Introduction

The discovery of physics beyond the Standard Model (BSM) at the Large Hadron Collider (LHC) typically relies on targeted searches optimised for a specific signal hypothesis. Model-agnostic *anomaly detection* approaches instead aim to identify rare, unusual events against the dominant quantum chromodynamics (QCD) background, without assuming a particular new-physics model.^{1,2} They are motivated by the possibility that the full LHC dataset contains signatures that existing analyses have overlooked. Such strategies have also moved from proposal to practice: ATLAS and CMS have reported dijet and related resonance searches based on weakly supervised, unsupervised and semi-supervised anomaly detection.^{3–7}

Machine learning offers a natural framework for anomaly detection. Approaches explored so far include autoencoders,^{8–13} normalising flows and other density estimators,^{14–16} and classification-based methods.^{17–20} More recently, boosted decision trees have been shown to outperform deep networks as classifiers in weakly supervised searches,²¹ and weak supervision has been combined with information from a class of signal models.²² The LHC Olympics (LHCO) 2020 challenge²³ provides a standardised benchmark with a hidden dijet resonance signal, enabling fair comparison across methods. The Dark Machines challenge²⁴ offers a complementary event-level benchmark, and resonant weakly supervised methods have been compared systematically on the LHCO R&D dataset.²⁵

Despite this activity, four questions remain under-explored. First, how sensitive are unsupervised detectors to accidental signal contamination in the training set? Second, what does a small set of labelled anomalies (weak supervision) add? Third, how much does each method sculpt the dijet mass spectrum, and does a mass-decorrelated architecture actually help? Fourth, do learned anomaly scores transfer to structurally different signal topologies?

Table 1. Summary of the anomaly detectors. Model IDs M0–M12, their supervision regime and anomaly score. ϕ is the network embedding and c the fixed hypersphere centre.

ID	Model	Supervision	Score
M0	τ_{21} cut	None	$-(\tau_{21}^I + \tau_{21}^J)$
M1	Autoencoder	Unsupervised	Reconstruction MSE
M2	Isolation Forest	Unsupervised	Mean isolation depth
M3	Deep SVDD	Unsupervised	$\ \phi(x) - c\ ^2$
M4	Deep SAD	Weak	$\ \phi(x) - c\ ^2, (\cdot)^{-1}$ for $y=1$
M5	Supervised MLP	Supervised	$\hat{p}(y=1 x)$
M6	Mass-aware SVDD	Unsupervised	$\ \phi(x) - c\ ^2 (+m_{JJ})$
M7	CWoLa Hunting	SR/SB labels	$\hat{p}(\text{SR} x)$
M8	ANODE (GMM)	Density-based	$-\log p_{\text{SB}}(x)$
M9	MD-SWAD	Unsupervised	$\ \phi(x)\ ^2$
M10	MD-SWAD-SAD	Weak	$\ \phi(x)\ ^2$
M11	CATHODE (CVAE)	SR/SB labels	$\hat{p}(\text{data} x)$
M12	CWoLa Trees	SR/SB labels	$\hat{p}(\text{SR} x)$

Here I address all four questions in a reproducible study on the LHCO 2020 R&D dataset. I reproduce and systematically compare six anomaly-detection architectures, with controlled randomisation over multiple seeds. I diagnose the collapse of Deep SVDD and a label-assignment error in Deep SAD, and apply targeted fixes. Mass sculpting and discrimination are characterised through the Jensen–Shannon divergence (JSD), background rejection at fixed signal efficiency, the maximum significance improvement characteristic and bump-hunt significance, and generalisation is tested on an unseen three-prong signal topology ($W' \rightarrow XY, X, Y \rightarrow qq$). I additionally implement a simple τ_{21} cut; resonant-search baselines (CWoLa Hunting, an ANODE-style Gaussian-mixture density estimator, a CATHODE-style proxy and a tree-based CWoLa variant); a mass-decorrelated sliced-Wasserstein detector with a semi-supervised extension; and an extended seven-feature dataset that adds τ_{32} for both jets. Follow-up studies test a constituent-level permutation-invariant model, an independent external signal, sideband score calibration and background-only closure.

Results

Benchmark overview

I use the LHCO 2020 R&D dataset,²³ which contains 1,000,000 simulated QCD dijet background events and 100,000 signal events from $Z' \rightarrow XY$ with $X \rightarrow q\bar{q}$ and $Y \rightarrow q\bar{q}$. Each event is described by five high-level jet-substructure features (Table 8 in Methods). Thirteen detectors, M0–M12, span the supervision spectrum from purely heuristic cuts to fully supervised classifiers (Table 1). The neural models share a bias-free linear-layer backbone, as the Deep SVDD family of losses requires.^{26–28} The headline comparison below concerns M1–M6; the additional detectors M0 and M7–M12 are described in Methods, and M7–M12 are compared in a dedicated subsection below.

Headline comparison

Table 2 reports the main performance metrics for the headline configuration of each model, averaged over seeds 42–46 for M1–M6. The fully supervised multilayer perceptron (M5) sets an oracle upper bound of $\text{AUC} = 0.9659$. Among the unsupervised methods in the headline set, Isolation Forest²⁹ (M2) performs best, with $\text{AUC} = 0.8068$, clearly outperforming the autoencoder and both SVDD variants. Deep SAD

Table 2. Headline results (mean $\pm$ s.d., five seeds for the six core models; single evaluation for deterministic cuts and specific baselines). Best unsupervised result in bold; oracle upper bound in italics. JSD is evaluated at 10% background efficiency.

Model	f_{train}	k	AUC	$1/\varepsilon_B$ at $\varepsilon_S =$			SIC _{max}	JSD (10%)	$\rho_{S,mJJ}$
				10%	30%	50%			
M1: Autoencoder	0%	0	0.718 ± 0.074	22.5	10.7	5.1	1.207 ± 0.191	0.0168	0.065
M2: Isolation Forest	0%	0	0.807 ± 0.008	29.0	13.1	7.8	1.521 ± 0.055	0.0287	0.285
M3: Deep SVDD	0%	0	0.500 ± 0.007	6.5	3.3	2.1	1.000 ± 0.000	0.0012	0.022
M4: Deep SAD	0.5%	100	0.888 ± 0.011	452.7	119.9	43.5	3.402 ± 0.433	0.0066	0.042
M5: Supervised	0%	1000	0.966 ± 0.000	39330.2	2189.6	402.0	25.128 ± 2.963	0.0082	0.016
M6: Mass-aware SVDD	0%	0	0.515 ± 0.010	7.6	3.5	2.2	1.000 ± 0.000	0.0262	0.071
M0: TauCut	0%	0	0.891	251.2	52.4	20.9	2.30	0.0032	0.062
M7: CWoLa	0%	0	0.605	10.5	4.0	2.6	1.07	0.1890	0.433
M8: ANODE	0%	0	0.720	18.1	7.5	4.3	1.20	0.0580	0.337
M11: CATHODE: CVAE	0%	0	0.481	1.2	1.2	1.2	1.00	0.0002	-0.028
M12: CWoLaTrees	0%	0	0.575	9.6	4.0	2.4	1.04	0.1887	0.451

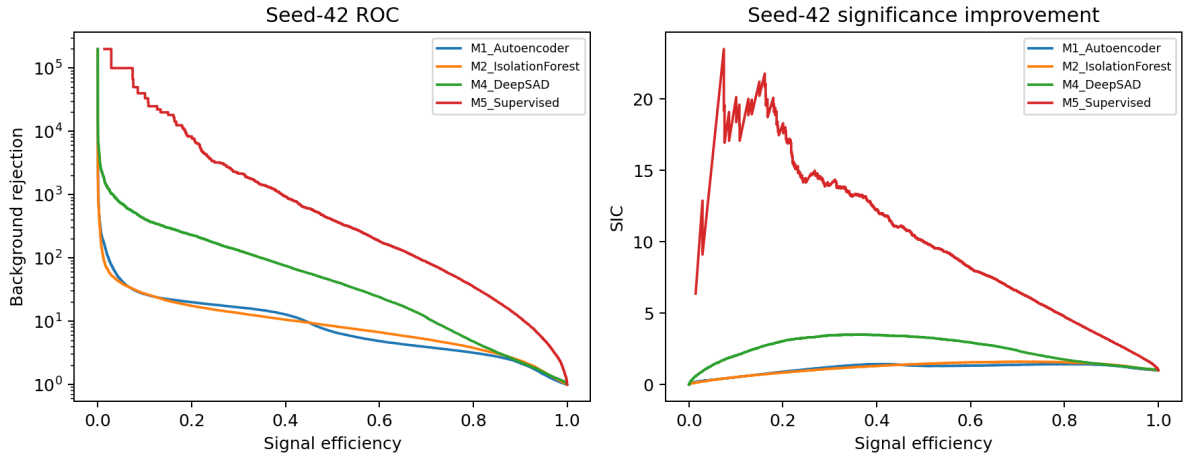


Fig. 1. ROC and SIC curves for the headline models. ROC curve (left, log-scale $1/\varepsilon_B$) and SIC curve (right) for the four informative headline models (seed 42). Deep SAD (M4) improves over the learned unsupervised baselines while remaining below the supervised reference (M5).

(M4) at $f_{\text{train}} = 0.5\%$ and $k = 100$ reaches $\text{AUC} = 0.8880$, recovering much of the gap between M2 and M5 with only 100 labelled examples. Fig. 1 shows the receiver-operating-characteristic (ROC) and significance improvement characteristic (SIC) curves for the four informative headline models.

Effect of weak supervision in Deep SAD

Table 3 gives the full (f_{train}, k) grid for Deep SAD, reported for seed 42, and Fig. 2 shows the same information as a heat map. Increasing the label budget improves the seed-42 ranking in each contamination regime. The contamination response is not monotonic: unlabelled injected signals can help or hurt at a fixed label budget. These single-seed sweep results do not identify a causal mechanism or establish an optimal label budget. The five-seed headline comparison holds the contamination and label budget fixed.

Effect of contamination on unsupervised models

Fig. 3 and Table 4 show how f_{train} affects the unsupervised detectors. Isolation Forest is robust across the full contamination range ($\text{AUC} \approx 0.81\text{--}0.82$). The autoencoder behaves non-monotonically, fluctuating significantly as contamination increases, without a clear trend. Deep SVDD degrades with f_{train} but stays

Table 3. Deep SAD ROC AUC over the contamination–label grid. AUC as a function of training contamination f_{train} and label budget k (seed 42).

f_{train}	$k = 10$	$k = 100$	$k = 1000$
0%	0.761	0.895	0.928
0.5%	0.845	0.876	0.932
1%	0.770	0.889	0.926

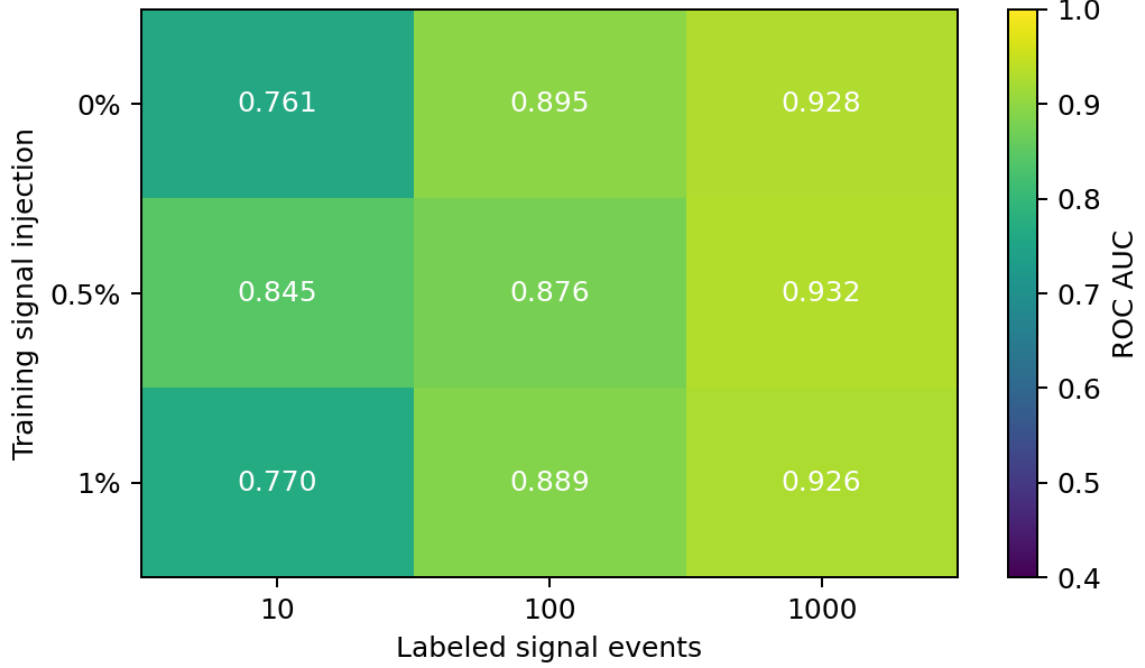


Fig. 2. Deep SAD AUC heat map. AUC over the (f_{train}, k) grid (seed 42).

near-random throughout.

Mass sculpting

Figs. 4 and 5 show three-panel m_{JJ} diagnostic plots for the two best-performing models; the corresponding plot for Isolation Forest is in Supplementary Fig. 1. Each figure shows (i) the normalised m_{JJ} spectrum before and after score cuts at 10% and 1% background efficiency; (ii) the bin-by-bin background acceptance as a function of m_{JJ} ; and (iii) the mean anomaly score per m_{JJ} bin, which reveals any residual score–mass correlation.

Fig. 6 plots AUC against JSD for all headline models across all seeds. The ideal detector lies in the lower-right corner (high AUC, low JSD). Deep SAD occupies a highly favourable position, combining near-oracle discrimination with a JSD comparable to that of the supervised model. Isolation Forest, although the best unsupervised detector in AUC, has the largest score–mass correlation ($\rho = 0.284$), which produces a non-flat acceptance profile that preferentially retains high-mass events (Supplementary Fig. 1). This underlines the importance of monitoring JSD alongside AUC.

Bump hunt

Table 5 reports the local significance Z before any cut, after a cut at 10% background efficiency and after

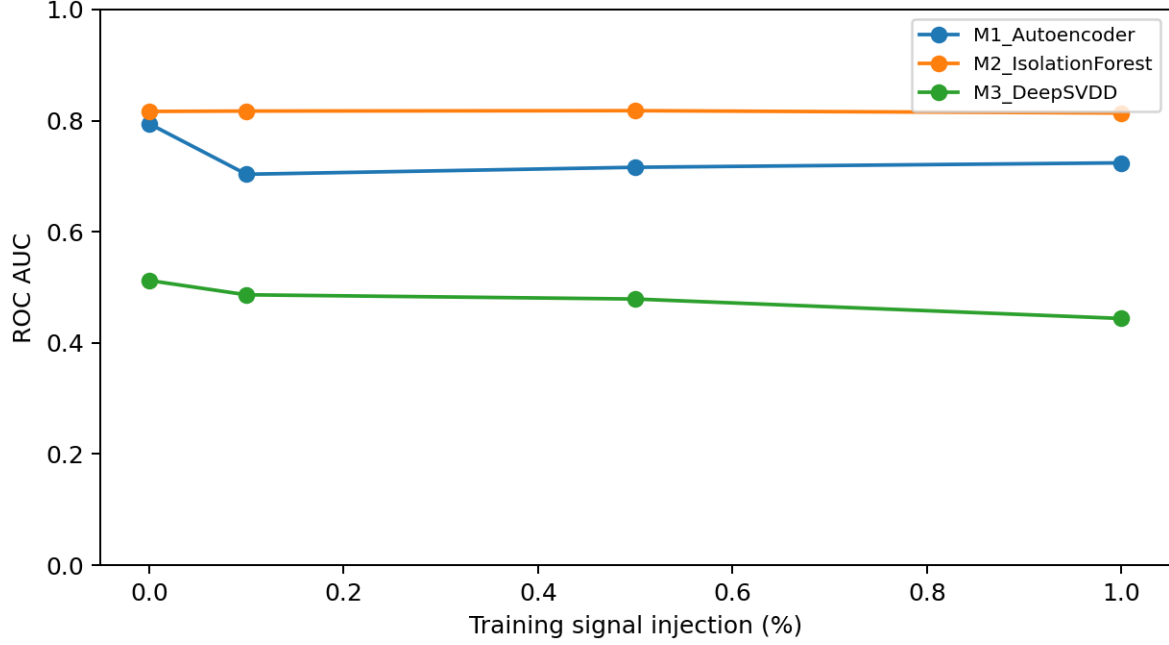


Fig. 3. Robustness of unsupervised detectors to contamination. ROC AUC versus training contamination f_{train} for the three unsupervised models (seed 42). Isolation Forest is robust; Deep SVDD degrades monotonically.

Table 4. Unsupervised AUC versus contamination. AUC as a function of f_{train} (seed 42).

f_{train}	M1: AE	M2: IF	M3: SVDD
0%	0.794	0.817	0.512
0.1%	0.703	0.817	0.487
0.5%	0.716	0.818	0.479
1%	0.724	0.814	0.444

a cut at 1% background efficiency, for two headline models. These are descriptive local fixed-window diagnostics on a simulated signal mixture. A small pre-cut excess does not establish calibrated false-positive control. No discovery significance, global search significance or detector-data sensitivity is claimed.

Failure modes

Deep SVDD hypersphere collapse. Established in this benchmark: Deep SVDD (M3) has near-random nominal-signal ranking under the released implementation.²⁶ **Plausible explanation:** low-variance embeddings and vanishing gradients in a multilayer bias-free network could contribute to this behaviour. The centre is initialised from training embeddings and its small components are clamped away from zero; it is not fixed to a uniform vector. No systematic centre/activation ablation record is included in this release. The numerical result therefore does not establish a universal architectural failure of Deep SVDD on tabular data.

Deep SAD label-assignment error. The original training script incorrectly assigned $y = 1$ (labelled anomaly) to all signal events injected as unlabelled contamination, rather than only to the k explicitly drawn labelled events. This gave Deep SAD spurious positive labels and created an artificial performance gain at $f_{\text{train}} > 0$, violating the assumptions of semi-supervised learning. After the fix, unlabelled injected

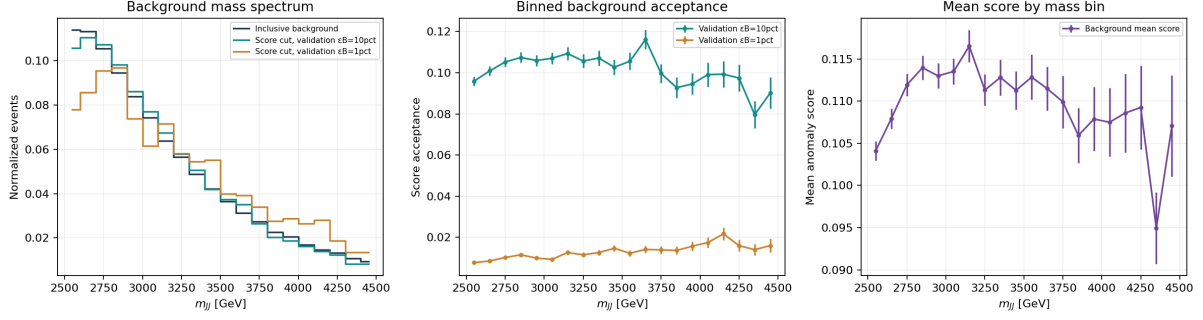


Fig. 4. Mass-sculpting diagnostics for Deep SAD. $f_{\text{train}} = 0.5\%$, $k = 100$, seed 42. The background m_{JJ} spectrum is largely preserved at 10% background efficiency (teal). A modest excess near 3.5 TeV appears at 1% efficiency (orange), consistent with the injected signal. The score–mass correlation is reported in the measured results table.

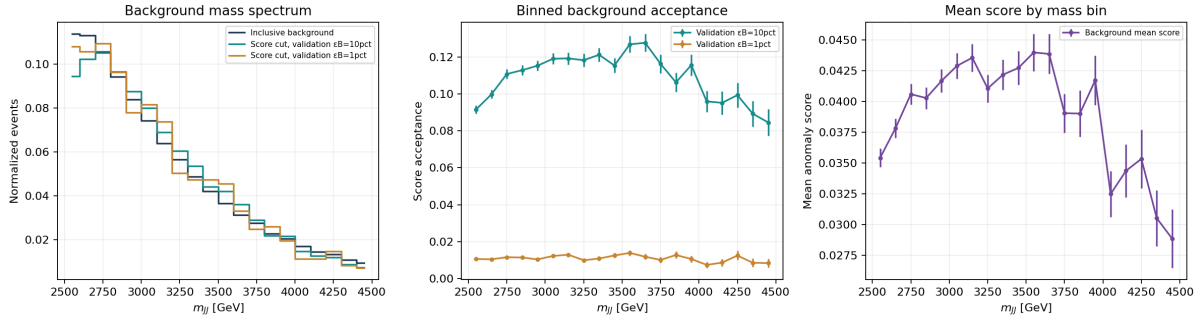


Fig. 5. Mass-sculpting diagnostics for the supervised MLP. $k = 1000$, seed 42. The 1% cut reveals a clear resonant excess near 3.5 TeV. The score–mass correlation is reported in the measured results table.

events correctly receive $y = 0$ (Eq. (3)), and the contamination benefit is attributed to the shift in the training-data distribution rather than to falsely labelled anomalies.

Mass-decorrelated and resonant-search baselines

The remaining detectors in Table 2 (M7–M12) address two of the questions posed in the Introduction: whether explicit mass decorrelation helps, and how resonant-search strategies compare with the substructure-based detectors.

The MD-SWAD implementations (M9 and M10) are exploratory extensions. Their earlier numerical summaries are [unverified]: no matching multi-seed evaluation table accompanies the saved checkpoints. They are excluded from the final quantitative comparison. Explicit decorrelation is therefore a candidate extension rather than an established result of this release.

The resonant-search baselines trail Isolation Forest (M2, $\text{AUC} = 0.8068$). ANODE with a Gaussian mixture (M8) reaches $\text{AUC} = 0.720$, CWoLa Hunting (M7) 0.605 and its tree-based variant (M12) 0.575, while the CATHODE (CVAE) proxy (M11) performs at chance level ($\text{AUC} = 0.481$, with $1/\epsilon_B = 1.2$ at every signal efficiency). M7, M8 and M12 also sculpt the mass spectrum strongly ($\text{JSD} = 0.189, 0.058$ and 0.189 ; $\rho = 0.433, 0.337$ and 0.450), consistent with their use of m_{JJ} -defined regions in training. Because M8, M11 and M12 are simplified proxies of the published methods (Methods), these numbers characterise my implementations rather than the methods themselves.

Generalisation to an unseen signal topology

I evaluate all headline models on the held-out three-prong signal sample ($X \rightarrow YY \rightarrow qq\bar{q}$) to assess

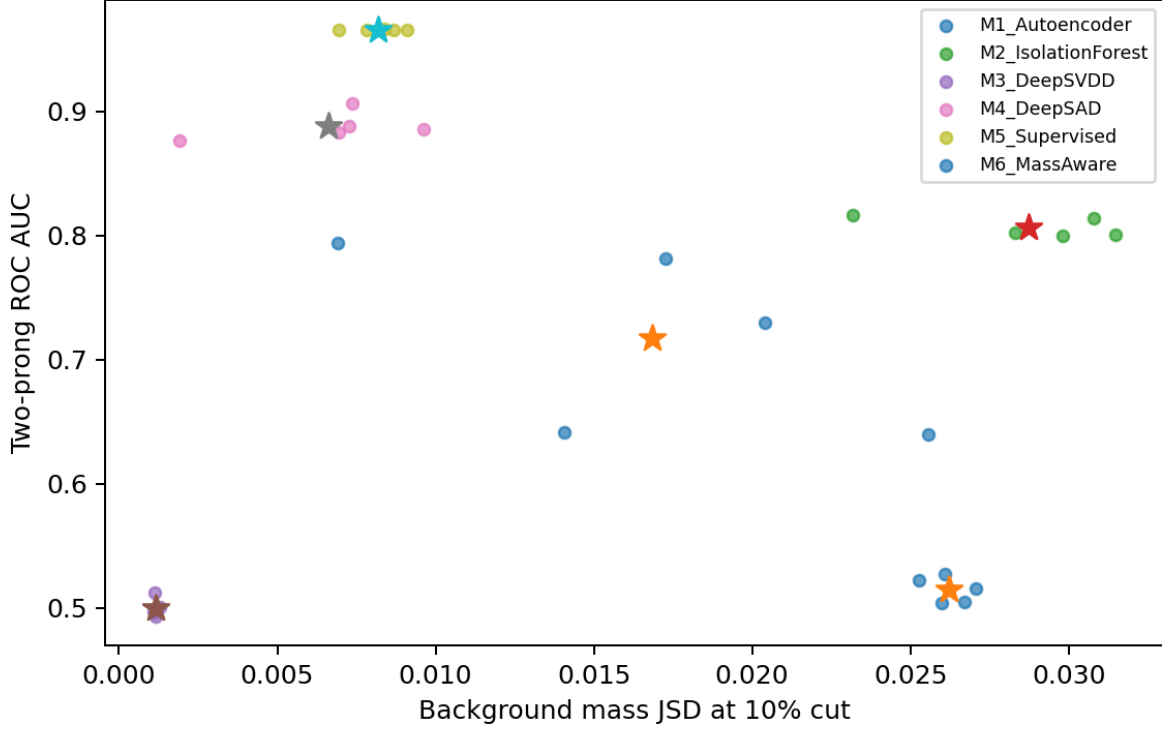


Fig. 6. Discrimination–sculpting trade-off. Each point is one (model, seed) pair; stars mark the per-model mean. Deep SAD (M4) is the semi-supervised method; its discrimination and mass-shape trade-off are measured directly rather than inferred from AUC alone.

generalisation beyond the two-prong training target. Fig. 7 shows the three-prong AUC against the two-prong AUC for all seeds and headline configurations, and Table 6 gives the quantitative comparison. The supervised MLP (M5) has the smallest generalisation gap (see the measured gaps in the table), suggesting that the learned τ_{21} and jet-mass discriminants partially transfer to the three-prong topology. Deep SAD (M4) shows a larger gap, indicating that sparse labelling is sufficient to learn partly topology-agnostic anomaly features.

Extended τ_{32} features. Plausible extension: τ_{32} measures three-subjettiness relative to two-subjettiness and may add information about three-prong substructure.⁴¹ The extended cache is available, but this release contains no matched five-seed comparison establishing an AUC improvement from these inputs.

Constituent-level representation and external signal

To test an event representation beyond the five- and seven-feature inputs, I trained a masked, permutation-invariant Deep Sets autoencoder on raw particle constituents. Each event contains up to 700 (p_T, η, ϕ) triples, transformed into six channels: $p_T / \sum p_T$, $\log(p_T / p_{T,\max})$, $\log(1 + p_T)$, scaled η , $\sin \phi$ and $\cos \phi$. The model was trained for eight epochs on a seeded subset of 50,000 events from the 599,952-event background training partition; the best validation-loss checkpoint was retained. A threshold for 1% validation background acceptance was fixed before testing. On the full original test pools, the two-prong and three-prong AUCs were 0.498 and 0.441, with signal efficiencies of 0.160% and 0.098% at that threshold. This run shows no useful separation from the constituent-level score.

I also evaluated all 44,673 events in the external $X \rightarrow YY' \rightarrow 4q$ sample (2+2-prong topology) released with additional LHCO signal models.⁵¹ Against a seeded 20,000-event subset of held-out background, the

Table 5. Bump-hunt local diagnostic. Local excess statistic Z from a binned Poisson sideband fit with propagated background-fit uncertainty, for Deep SAD and the supervised MLP. Signal injection is 0.5% of the background count, corresponding to approximately 0.498% mixture prevalence.

Model	Seed	Pre-cut	@ 10% eff.	@ 1% eff.
M4: Deep SAD	42	0.00	5.89	7.58
M4: Deep SAD	43	0.00	4.71	5.80
M4: Deep SAD	44	0.03	4.13	invalid fit
M4: Deep SAD	45	0.00	4.95	7.51
M4: Deep SAD	46	0.00	4.87	8.43
M5: Supervised	42	0.00	6.63	13.69
M5: Supervised	43	0.00	6.66	14.71
M5: Supervised	44	0.03	7.76	14.86
M5: Supervised	45	0.00	6.59	13.76
M5: Supervised	46	0.00	6.32	13.96

Table 6. Generalisation study. Two-prong AUC (training target) versus three-prong AUC (held out, never seen during training), as five seeds 42–46. Generalisation gap = $\text{AUC}_{3p} - \text{AUC}_{2p}$. *A positive gap is accidental: SVDD collapse produces a near-random two-prong score that happens to correlate with three-prong substructure.

Model	AUC (2-prong)	AUC (3-prong)	Gap
M1: Autoencoder	0.718 ± 0.074	0.597 ± 0.103	-0.121
M2: Isolation Forest	0.807 ± 0.008	0.683 ± 0.021	-0.124
M3: Deep SVDD	0.500 ± 0.007	0.617 ± 0.011	$+0.117$
M4: Deep SAD	0.888 ± 0.011	0.749 ± 0.061	-0.139
M5: Supervised	0.966 ± 0.000	0.929 ± 0.001	-0.037
M6: Mass-aware SVDD	0.515 ± 0.010	0.667 ± 0.023	$+0.152$

AUC was 0.465 and the signal efficiency at the frozen threshold was 0.159%. This sample was not used in training or threshold selection, and its generator chain differs from the original R&D benchmark. It is therefore reported as an external evaluation, not pooled with the nominal benchmark.

The second external $W_{KK} \rightarrow WR \rightarrow 3W$ sample has a 2+4-prong topology. Its archive matches the published MD5 checksum; all 32,801 events were scored with the same frozen checkpoint. Against the same seeded 20,000-event held-out background subset, the AUC was 0.507 and the signal efficiency at the frozen 1% threshold was 0.280%. This near-random result is reported separately because the generation chain differs from the original R&D benchmark.

Density-score calibration and background-only closure

For the ANODE-style GMM, 128,112 held-out background events in the mass sidebands were split into independent calibration and evaluation halves of 64,056 events each. I converted the negative log-density score s to an upper-tail conformal p -value, $p = (1 + \#\{s_i^{\text{cal}} \geq s\}) / (n_{\text{cal}} + 1)$. Under exchangeability of the calibration and evaluation background, this gives finite-sample calibrated score p -values.⁵² The evaluation values were consistent with uniformity (KS statistic 0.0043, $p = 0.179$); observed rates below 1%, 5% and 10% were 1.054%, 4.975% and 9.868%. This calibrates the anomaly-score tail for sideband background, not an absolute density or a signal-region conditional density. Consistent with this distinction, the raw GMM Rosenblatt-PIT marginals remained nonuniform on the evaluation half (KS statistics 0.026–0.052, all $p < 10^{-38}$).

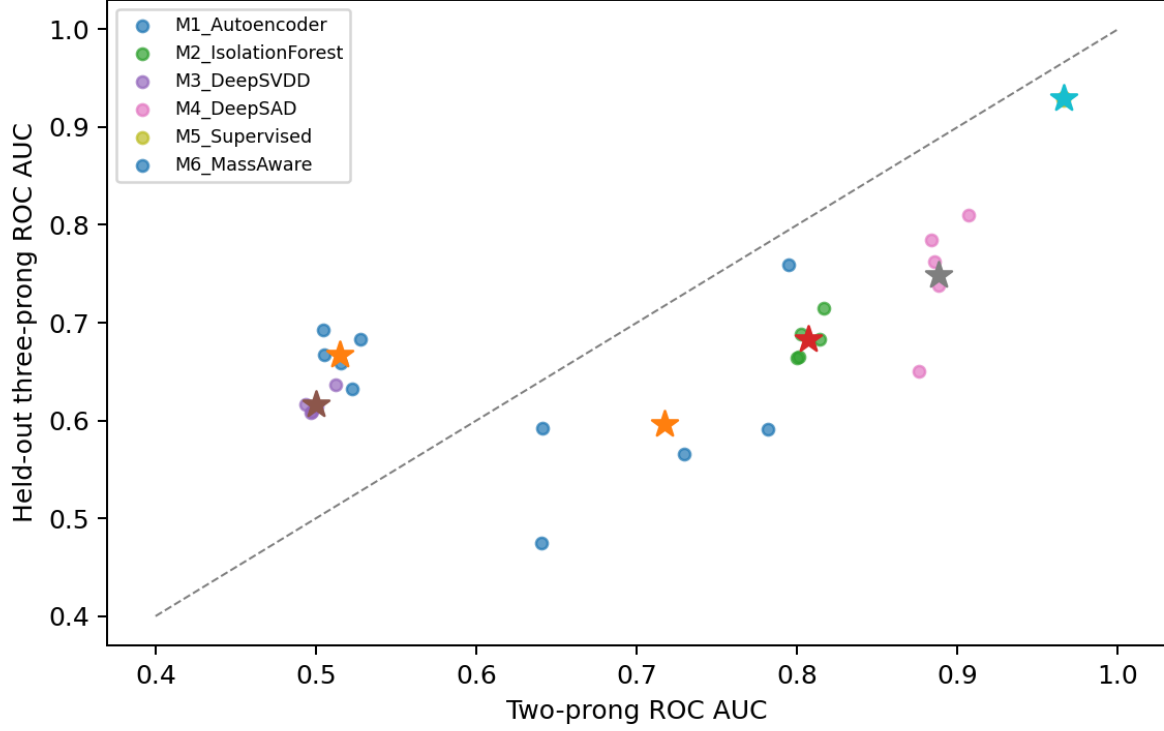


Fig. 7. Generalisation to the unseen three-prong topology. Points above the dashed diagonal generalise better to three-prong than to two-prong signal. Deep SAD (M4) maintains the best combination of two-prong and three-prong AUC among the unsupervised and semi-supervised methods.

As a background-only closure check, I generated 500 Poisson-bootstrap pseudoexperiments from held-out background, froze each detector threshold using validation background, and repeated the local sideband fit. At the 1% operating point, the 95th-percentile local significances were 1.86 (M1), 1.19 (M2) and 1.51 (M11); the fractions with $Z \geq 3$ were 0/461, 0/479 and 1/456 among valid fits. These are local conditional closure results for the released simulation and chosen fit, not global significance estimates or detector-systematic closure.

Discussion

I have presented a benchmark of unsupervised and semi-supervised anomaly detection on the LHC0 2020 R&D dataset. Weak supervision is highly label-efficient: with $k = 100$ labelled anomalies, Deep SAD achieves $AUC = 0.8880$, recovering much of the AUC gap between the best unsupervised baseline (Isolation Forest, $AUC = 0.807 \pm 0.008$) and the supervised reference ($AUC = 0.9659$). Its fixed-window local diagnostics on the simulated signal mixture are reported separately from its measured mass sculpting ($JSD = 0.0066$). Isolation Forest is the best learned unsupervised detector on this five-dimensional feature set and is robust to training contamination. Deep SVDD, by contrast, collapses on zero-centred tabular data in this implementation. The mechanism remains uncertain and the limited implementation comparison warrants caution when applying SVDD to standardised tabular data in high-energy physics. Generalisation to an unseen three-prong topology is partial but meaningful: Deep SAD retains $AUC = 0.7492$ without any three-prong training events.

These results offer practical guidance, though they must be interpreted within the scope of a simulation-

only benchmark. The headline comparisons use low-dimensional features and a nominal signal; follow-up tests add one external signal and a constituent-level model but do not establish broad signal-model coverage. Within the headline regime, Isolation Forest is a sensible unsupervised baseline: it needs no signal assumption, trains in minutes and is robust to contamination up to 1%. Deep SAD should be preferred over Deep SVDD for similar tabular feature sets, because it performs better in this measured setup; this is not a claim that the repulsion term removes all degenerate fixed points. For the two-prong signal studied here, 100 labelled examples improve on the learned unsupervised baseline, while the seed-42 grid shows additional gains with 1,000 labels. A repeated-seed grid would be needed to establish the trade-off between label cost and performance.

Mass sculpting must be evaluated carefully before any score cut is deployed. Deep SAD and Isolation Forest showed acceptable JSD (< 0.03) at 10% background efficiency in this five-feature space, but the autoencoder and mass-aware SVDD would require additional decorrelation in any production search, such as distance correlation³⁰ or adversarial training.^{31;32} The exploratory MD-SWAD implementations require a verified repeated-seed evaluation before their discrimination–decorrelation trade-off can be quantified. Resonant methods designed to avoid background sculpting are discussed in Ref. 33. The seven-feature τ_{32} dataset remains a candidate for direct high-level model evaluation. The separate constituent-level Deep Sets test reported here did not improve discrimination, showing that higher input dimensionality alone is insufficient.

In principle, CWoLa Hunting is complementary to the substructure-based scores, because it uses the m_{JJ} window as a physics-motivated weak label. In this study, however, the simplified resonant-search baselines (M7, M8, M11 and M12) all trailed Isolation Forest and, for M7, M8 and M12, sculpted the mass spectrum strongly. They should be revisited with full implementations, such as normalising-flow density estimators, before any conclusion is drawn about the published methods. The original M0–M12 comparison uses five- or seven-feature inputs; the follow-up Deep Sets model instead processes up to 700 constituents with six channels per constituent, but its AUCs remain near or below chance. These results provide a low-level reference point alongside the high-level baselines for future collider anomaly-detection studies, including emerging quantum approaches.^{34;35}

Methods

Dataset. The LHCO 2020 R&D dataset^{23;36} consists of 1,100,000 dijet events: 1,000,000 QCD background events produced with Pythia 8³⁷ and Delphes,³⁸ and 100,000 signal events from the process $W' \rightarrow XY$, $X \rightarrow q\bar{q}$ and $Y \rightarrow q\bar{q}$, at $m_{W'} = 3.5$ TeV, $m_X = 500$ GeV and $m_Y = 100$ GeV. The signal produces a characteristic two-pronged topology in each of the two leading jets. A second, held-out signal sample with a three-prong topology ($W' \rightarrow XY$, $X, Y \rightarrow qqq$, with the same particle masses) is used exclusively for the generalisation evaluation. Events are split into disjoint training, validation and test sets (Table 7). Training-set signal events are used only when explicitly injected (contamination studies) or labelled (Deep SAD and supervised models).

Feature extraction. High-level features are derived from the two leading anti- k_T jets³⁹ ($R = 1.0$), using FastJet⁴⁰ and the official FastJet-contrib N-subjettiness implementation.⁴¹ The five standard features used by all models are listed in Table 8. The dijet invariant mass m_{JJ} is deliberately withheld from the standard feature set to prevent trivial mass sculpting; it is used only by mass-aware SVDD (M6), CWoLa (M7, M12), ANODE (M8), MD-SWAD (M9, M10), and CATHODE (M11). All features are standardised (zero mean, unit variance) with a `StandardScaler`⁴² fitted on the background training set only, never on

Table 7. Dataset splits. Event counts used throughout this study.

Split	Background	Signal (2-prong)
Train	599,952	59,983
Validation	199,984	19,994
Test	199,984	19,996

Table 8. Standard five-feature set. Feature indices 0–4.

Idx	Feature	Description
0	m_{J_1}	Leading-jet invariant mass [GeV]
1	$ \Delta m_J $	Jet-mass asymmetry $ m_{J_1} - m_{J_2} $ [GeV]
2	τ_{21}^J	2-to-1 N-subjettiness, leading jet
3	$\tau_{21}^{J_2}$	2-to-1 N-subjettiness, sub-leading jet
4	ΔR_{JJ}	Angular separation of the two jets

signal events.

Extended feature set. For the generalisation study, I additionally compute the 3-to-2 N-subjettiness ratio $\tau_{32}^{J_{1,2}} = \tau_3/\tau_2$ for both jets, extending the feature vector to seven dimensions. This variable directly discriminates three-prong decay products from QCD dijets. The extended dataset is stored at `data/processed/events_v2_extended_features.h5`.

Raw-particle extension. The separate constituent-level study reads up to 700 raw (p_T, η, ϕ) triples per event and maps each particle to six channels: $p_T/\sum p_T$, $\log(p_T/p_{T,\max})$, $\log(1+p_T)$, scaled η , $\sin \phi$ and $\cos \phi$. A shared particle encoder is followed by masked mean and maximum pooling, an event embedding, and a per-particle decoder. The event anomaly score is the mean squared reconstruction error over valid particles; it is a ranking score, not a likelihood. Training uses background only: 50,000 seeded events from a 599,952-event training partition for eight epochs. The checkpoint with the lowest validation loss is kept, and the score threshold is fixed at the validation 99th percentile with randomized acceptance at tied scores. All original test pools are scored. The additional $X \rightarrow YY' \rightarrow 4q$ sample contains 44,673 events and comes from Zenodo record 18983506;⁵¹ its AUC is measured against a seeded 20,000-event subset of held-out background. The external $W_{KK} \rightarrow WR \rightarrow 3W$ sample from the same record contains 32,801 events and is evaluated with the same checkpoint, background subset and threshold.

Conformal density-score calibration. For the ANODE-style GMM, held-out background events in the 2.5–3.3 and 3.7–4.5 TeV sidebands are divided into disjoint calibration and evaluation halves. For evaluation score s , the upper-tail conformal p -value is $(1 + \#\{s_i^{\text{cal}} \geq s\})/(n_{\text{cal}} + 1)$, with ties counted conservatively. Its finite-sample validity requires exchangeability between these two background halves.⁵² The resulting values calibrate the score tail for sideband background only; they are not an absolute density estimate and do not establish transfer to the signal region.

Background-only closure. For M1, M2 and M11, validation-background thresholds are frozen before generating 500 Poisson-bootstrap pseudoexperiments from held-out background. Score ties at the cut use randomized boundary acceptance. The local sideband fit is repeated for each trial; failed or saturated fits are excluded, so the valid count is reported for each model and working point. This conditional simulation closure does not include detector systematics, alternative background generators or a global search.

Model architectures. SVDD and SAD use bias-free layers with architecture $5 \rightarrow 64 \rightarrow 32 \rightarrow 16$. The

autoencoder has an encoder $5 \rightarrow 64 \rightarrow 32 \rightarrow 16$ and a symmetric decoder, with biases and LeakyReLU activations. The supervised reference uses biased layers $5 \rightarrow 64 \rightarrow 32 \rightarrow 1$ with ReLU activations. The mass-aware model has six inputs; the hidden widths are otherwise unchanged.

Loss functions. The autoencoder (M1) minimises the mean-squared reconstruction error,

$$\mathcal{L}_{\text{AE}} = \frac{1}{n} \sum_{i=1}^n \|x_i - \hat{x}_i\|^2. \quad (1)$$

Deep SVDD²⁶ (M3) minimises the mean squared distance from a fixed centre c ,

$$\mathcal{L}_{\text{SVDD}} = \frac{1}{n} \sum_{i=1}^n \|\phi(x_i) - c\|^2. \quad (2)$$

Deep SAD²⁷ (M4) extends this with an inverse-distance penalty for labelled anomalies ($y_i = 1$),

$$\mathcal{L}_{\text{SAD}} = \frac{1}{n} \sum_{y_i=0} \|\phi(x_i) - c\|^2 + \frac{\eta}{n} \sum_{y_i=1} (\|\phi(x_i) - c\|^2)^{-1}, \quad (3)$$

where $\eta > 0$ controls the strength of the anomaly repulsion. Unlabelled events ($y_i = 0$) include both background training events and any signal events injected without labels. To resolve the Deep SVDD collapse and explicitly decorrelate the latent representation from the dijet mass m_{JJ} , I propose the Mass-Decorrelated Sliced Wasserstein Anomaly Detector (MD-SWAD, M9). MD-SWAD maps the background directly onto a standard normal target distribution using the sliced Wasserstein distance (SWD),^{43,44} and adds a distance-correlation (DisCo) penalty^{30,45}, weighted by λ , that decorrelates the embedding from m_{JJ} :

$$\mathcal{L}_{\text{MD-SWAD}} = \text{SWD}(\phi(X_{y=0}), \mathcal{N}(0, I)) + \lambda \cdot \text{dCor}^2(\phi(X_{y=0}), m_{JJ}). \quad (4)$$

For the semi-supervised variant (MD-SWAD-SAD, M10), the Deep SAD repulsion term for the labelled anomalies ($y_i = 1$) is added to push them away from the origin:

$$\mathcal{L}_{\text{MD-SWAD-SAD}} = \mathcal{L}_{\text{MD-SWAD}} + \frac{\eta}{n} \sum_{y_i=1} (\|\phi(x_i)\|^2)^{-1}. \quad (5)$$

Training protocol. All neural models are implemented in PyTorch⁴⁶ and trained with Adam⁴⁷ (learning rate 10^{-3} , weight decay 10^{-5}), a batch size of 256 and early stopping with patience 8 on the validation loss. The checkpoint with the lowest validation objective is selected. Five seeds (42–46) are used for the six core headline comparisons; supplemental baselines are single-seed exploratory evaluations. The final core results use 47 fresh fits in an isolated Python 3.11 CPU environment; dependency versions and the staged 33-minute verification are recorded in the repository. Historical neural results differ and are preserved separately. For the final seed-42 detectors, 500 stratified paired-bootstrap trials on 199,984 background and 19,996 signal test events provide separate AUC intervals in the released machine-readable tables. These finite-test intervals are conditional on the fitted models and are not combined with training-seed standard deviations.

Contamination and label budget. The training contamination f_{train} is the number of unlabelled signal events injected into the training set, expressed as a fraction of the background training pool size ($N_{\text{bkg}} = 599,952$). For example, $f_{\text{train}} = 0.5\%$ adds 2,999 signal events to the training set without labels. The label budget k (Deep SAD only) is the number of distinct signal events assigned a positive label

($y = 1$) from the remaining training-signal pool. These events are drawn after the contamination injection and are mutually exclusive with the unlabelled injected events.

Evaluation metrics. I report the area under the receiver-operating-characteristic curve (AUC) together with the background rejection $1/\epsilon_B$ at signal efficiencies $\epsilon_S \in \{10\%, 30\%, 50\%\}$. The maximum significance improvement characteristic is

$$\text{SIC}_{\max} = \max_t \frac{\epsilon_S(t)}{\sqrt{\epsilon_B(t)}}. \quad (6)$$

Mass sculpting is quantified by the Jensen–Shannon divergence

$$\text{JSD}(P\|Q) = \frac{1}{2}D_{\text{KL}}(P\|M) + \frac{1}{2}D_{\text{KL}}(Q\|M), \quad (7)$$

where $M = (P + Q)/2$, P is the normalised inclusive background m_{JJ} distribution and Q is the normalised m_{JJ} distribution after a score cut at a fixed background efficiency; $\text{JSD} = 0$ means no sculpting. I also report the Pearson correlation coefficient $\rho(s, m_{JJ})$ between anomaly scores s and m_{JJ} on the background test sample, for which $|\rho| \rightarrow 0$ is ideal. The bump-hunt significance is the local excess statistic Z with propagated background-fit uncertainty. The background expectation comes from a binned Poisson fit to sidebands in 2.5–4.5 TeV, excluding the fixed 3.3–3.7 TeV window. The log-background model uses the basis 1, $\log(1 - x)$, $-\log x$ and $-(\log x)^2$, with $x = m_{JJ}/(13 \text{ TeV})$. No Gaussian signal template is fitted. Definitions of all metrics are collected in Supplementary Table 1.

Additional baselines. CWoLa Hunting^{17–19} (M7; Classification Without Labels) uses the m_{JJ} distribution to define a signal region (SR: 3300–3700 GeV) and sidebands (SB: 2500–3300 and 3700–4500 GeV). A classifier trained to separate SR from SB events learns to isolate signal (preferentially in the SR) from background, using m_{JJ} as a surrogate label rather than a direct anomaly tag. The approach is model-agnostic and requires no signal simulation. My implementation uses a compact MLP ($5 \rightarrow 64 \rightarrow 64 \rightarrow 1$, ReLU activations, BCEWithLogitsLoss) trained only on events within the SR+SB window, and is activated by the configuration flag `use_mjj: true` in `m7_cwola.yaml`. ANODE¹⁴ (M8) estimates the background density from sideband events and scores anomalies by the likelihood ratio $p_{\text{SR}}(x)/p_{\text{SB}}(x)$. My implementation approximates the sideband density with a five-component Gaussian mixture model (GMM) fitted on $x \in \text{SB}$ and uses the score $-\log p_{\text{SB}}(x)$ as a proxy for the likelihood ratio. A full ANODE implementation with normalising flows⁴⁸ would give a more accurate density estimate, particularly near the SR boundary.

CATHODE¹⁶ (M11) normally replaces the sideband density model with flow-based sampling of the background in the signal region, and then trains a classifier to distinguish data from the synthetic background. Because a full implementation requires a normalising flow, I implemented a computationally tractable proxy, CATHODE (CVAE), in which a conditional variational autoencoder^{49;50} generates the pseudo-background in place of the flow. Finally, CWoLa Trees (M12) follows the same weak-supervision methodology as M7 but replaces the neural network with a histogram-based gradient-boosting classifier,⁴² a standard tree-based weakly supervised baseline.²¹ The τ_{21} -cut baseline (M0) scores events by $-(\tau_{21}^{J_1} + \tau_{21}^{J_2})$.

Data availability

The LHC Olympics 2020 R&D dataset is publicly available from the challenge organisers.^{23;36} The external $X \rightarrow YY' \rightarrow 4q$ and $W_{KK} \rightarrow WR \rightarrow 3W$ signal samples used in the follow-up evaluations are available from Zenodo record 18983506.⁵¹ Derived feature files and results supporting this study are available at the repository given below.

Code availability

All code, configurations and results are available in the [public project repository](#).

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Author contributions

M.H.A. is the sole author of this work. M.H.A. conceived the study, developed the software, performed the experiments and analysis, interpreted the results, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Additional information

Supplementary information accompanies this paper.

Correspondence and requests for materials should be addressed to Mithil Hardik Astik.

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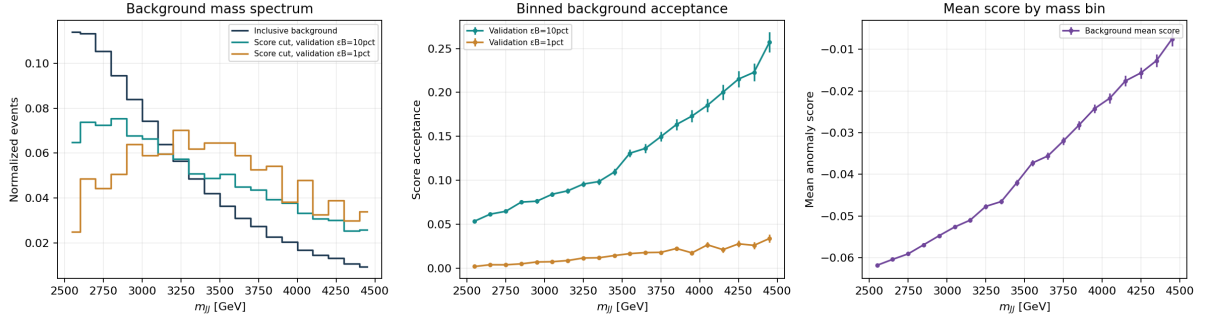
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Supplementary information



Supplementary Fig. 1. Mass-sculpting diagnostics for Isolation Forest. Seed 43. The score- m_{JJ} correlation The measured score-mass correlation is the largest among all headline models, leading to a mass-dependent acceptance profile even at moderate (10%) background efficiency.

Supplementary Table 1. Summary of evaluation metrics.

Metric	Definition
AUC	$\int_0^1 \varepsilon_S(\varepsilon_B) d\varepsilon_B$
$\text{SIC}_{\max}$	$\max_t \varepsilon_S(t) / \sqrt{\varepsilon_B(t)}$
$1/\varepsilon_B @ \varepsilon_S^*$	$1/\text{FPR}(t^*)$ where $\text{TPR}(t^*) = \varepsilon_S^*$
JSD	$\frac{1}{2}D_{\text{KL}}(P\ M) + \frac{1}{2}D_{\text{KL}}(Q\ M)$, $M = \frac{P+Q}{2}$; P, Q = inclusive/cut m_{JJ} histograms
$\rho_{s,m_{JJ}}$	Pearson correlation between scores and m_{JJ} on the background test set
f_{train}	Fraction of N_{bkg} injected as unlabelled signal
k	Number of positively labelled signal events (Deep SAD only)